

07 Feature Store

Course Name: Data Management for Machine Learning
Assignment Title: End-to-End Data Management Pipeline for a Recommendation System
Assignment: Group 51 - Data management for Machine Learning Group 51

Team Members

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1. Objective

This report documents the feature store stage of the recommendation pipeline. It captures the feature store configuration, metadata registry, versioned storage artifacts, and sample feature retrieval for both training and inference.

2. Objective Coverage

- Implement a simple custom feature store using a metadata registry and SQLite offline store.
- Document feature names, data sources, and transformation logic.
- Support versioned retrieval for both training and inference workflows.
- Capture project logs and evidence needed for a submission-ready PDF.

3. Supporting Project and Log Artifacts

The following discovered files support this section, including source datasets, validation outputs, feature store artifacts, and retrieval demos. Long file paths are wrapped inside table cells using real line breaks only at safe separators.

Artifact	Relative Path	Last Modified	Size
events.csv	data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv	2026-04-20 17:28:00	92029.21 KB
prepared_interactions_20260429_215532.parquet	data\prepared\prepared_interactions_20260429_215532.parquet	2026-04-29 21:55:34	65323.8 KB
products.parquet	data\bronze\dummyjson\products.parquet	2026-04-29 16:42:35	73.89 KB
run_validation.txt	src\02-validation\run_validation.txt	2026-04-28 20:14:12	46.94 KB
data_quality_report_20260429T111342Z.json	reports\validation\data_quality_report_20260429T111342Z.json	2026-04-29 16:47:43	6.66 KB
data_quality_report_20260429T111342Z.pdf	reports\validation\data_quality_report_20260429T111342Z.pdf	2026-04-29 16:47:44	8.49 KB

Artifact	Relative Path	Last Modified	Size
fix_log_20260429T111342Z.csv	reports\validation\fix_log_20260429T111342Z.csv	2026-04-29 16:46:38	3.43 KB
validation_summary_revalidated_20260429T111342Z.csv	reports\validation\validation_summary_revalidated_20260429T111342Z.csv	2026-04-29 16:47:43	0.41 KB
validation_issues_revalidated_20260429T111342Z.csv	reports\validation\validation_issues_revalidated_20260429T111342Z.csv	2026-04-29 16:47:43	0.51 KB
validation_log_20260429T111342Z.jsonl	logs\validation_log_20260429T111342Z.jsonl	2026-04-29 16:47:44	0.66 KB
feature_store.db	feature_store\offline_store\feature_store.db	2026-04-29 22:17:40	609080.0 KB
feature_store_config.json	feature_store\registry\feature_store_config.json	2026-04-29 22:17:42	2.25 KB
feature_metadata.csv	feature_store\registry\feature_metadata.csv	2026-04-29 22:17:42	3.7 KB
feature_metadata.json	feature_store\registry\feature_metadata.json	2026-04-29 22:17:42	7.91 KB
feature_store_schema.sql	feature_store\registry\feature_store_schema.sql	2026-04-29 22:17:42	1.25 KB
sample_training_feature_retrieval.csv	feature_store\docs\sample_training_feature_retrieval.csv	2026-04-29 22:17:43	0.67 KB
sample_inference_feature_retrieval.csv	feature_store\docs\sample_inference_feature_retrieval.csv	2026-04-29 22:17:43	0.42 KB
feature_store_log_20260429T164623Z.jsonl	logs\feature_store_log_20260429T164623Z.jsonl	2026-04-29 22:17:43	1.69 KB

4. Feature Store / Registry Assets in Project

Feature Store Asset	Relative Path	Last Modified	Size
feature_store_config.json	feature_store\registry\feature_store_config.json	2026-04-29 22:17:42	2.25 KB
feature_store_schema.sql	feature_store\registry\feature_store_schema.sql	2026-04-29 22:17:42	1.25 KB
feature_store_20260429T072423Z.json	logs\feature_store_20260429T072423Z.json	2026-04-29 13:18:24	0.18 KB
registry_loader_failed_20260429_160933.json	logs\monitoring\registry_loader_failed_20260429_160933.json	2026-04-29 16:26:06	3.54 KB
registry_loader_started_20260429_160933.json	logs\monitoring\registry_loader_started_20260429_160933.json	2026-04-29 16:26:00	0.41 KB
registry_loader_started_20260429_164131.json	logs\monitoring\registry_loader_started_20260429_164131.json	2026-04-29 17:05:20	0.41 KB
registry_loader_success_20260429_164131.json	logs\monitoring\registry_loader_success_20260429_164131.json	2026-04-29 17:05:35	0.41 KB
materialize_features_executed.ipynb	runs\prefect\20260429_160933\executed_notebooks\materialize_features\materialize_features_executed.ipynb	2026-04-29 16:25:43	17.27 KB
registry_loader_executed.ipynb	runs\prefect\20260429_160933\executed_notebooks\registry_loader\registry_loader_executed.ipynb	2026-04-29 16:26:06	21.42 KB

Feature Store Asset	Relative Path	Last Modified	Size
materialize_features_executed.ipynb	runs\prefect\20260429_164131\executed_notebooks\materialize_features_executed.ipynb	2026-04-29 17:05:20	17.27 KB
registry_loader_executed.ipynb	runs\prefect\20260429_164131\executed_notebooks\registry_loader\registry_loader_executed.ipynb	2026-04-29 17:05:35	28.02 KB
materialize_features.ipynb	src\04-features\materialize_features.ipynb	2026-04-29 15:22:54	15.82 KB
registry_loader copy.ipynb	src\04-features\registry_loader copy.ipynb	2026-04-28 20:31:57	3.61 KB
registry_loader.ipynb	src\04-features\registry_loader.ipynb	2026-04-29 16:38:41	26.43 KB
06 Feature Engineering and Transformation.ipynb	src\07-submission\06 Feature Engineering and Transformation.ipynb	2026-04-29 22:08:33	55.48 KB
07 Feature Store.ipynb	src\07-submission\07 Feature Store.ipynb	2026-04-29 22:16:26	74.69 KB
recommendation_features_schema.sql	warehouse\recommendation_features_schema.sql	2026-04-29 22:07:21	1.09 KB

5. Feature Store Logic Summary

- Loaded feature tables from existing transformed data or warehouse database.
- Feature store design uses a custom metadata registry instead of Feast.
- Offline store persists versioned feature tables in SQLite.
- Configuration is written to feature_store/registry/feature_store_config.json.
- Feature metadata is written to CSV and JSON registry files.
- Versioned retrieval uses *_v1 tables inside the feature store database.
- Training retrieval joins user, item, and user-item feature views.
- Inference retrieval demonstrates point-style feature lookup for a sample user and candidate items.

6. Versioned Feature Store Outputs

Output	Relative Path	Type	Last Modified	Size
feature_store.db	feature_store\offline_store\feature_store.db	.db	2026-04-29 22:17:40	609080.0 KB
feature_store_config.json	feature_store\registry\feature_store_config.json	.json	2026-04-29 22:17:42	2.25 KB
feature_metadata.csv	feature_store\registry\feature_metadata.csv	.csv	2026-04-29 22:17:42	3.7 KB
feature_metadata.json	feature_store\registry\feature_metadata.json	.json	2026-04-29 22:17:42	7.91 KB
feature_store_schema.sql	feature_store\registry\feature_store_schema.sql	.sql	2026-04-29 22:17:42	1.25 KB
sample_training_feature_retrieval.csv	feature_store\docs\sample_training_feature_retrieval.csv	.csv	2026-04-29 22:17:43	0.67 KB

Output	Relative Path	Type	Last Modified	Size
sample_inference_feature_retrieval.csv	feature_store\docs\sample_inference_feature_retrieval.csv	.csv	2026-04-29 22:17:43	0.42 KB

7. Feature Table Summary

table_name	rows	columns	duplicate_rows
user_features	1407580	5	0
item_features	235061	9	0
user_item_features	2145179	8	0
item_cooccurrence	32411667	3	0

8. Feature Store Version Summary

feature_store_version	created_ts	offline_store	training_demo_rows	inference_demo_rows
v1	2026-04-29T16:46:23Z	feature_store\offline_store\feature_store.db	10	10

9. Feature Metadata Documentation

The metadata registry below documents feature view, entity keys, feature name, data type, source path, and version.

feature_view	entity_keys	feature_name	dtype	source_path	version
item_cooccurrence	item_id_a, item_id_b	cooccurrence_count	int64	data\transformed\item_cooccurrence.csv	v1
item_features	item_id	avg_interaction_weight	float64	data\transformed\item_features.csv	v1
item_features	item_id	avg_rating_per_item	float64	data\transformed\item_features.csv	v1
item_features	item_id	category	object	data\transformed\item_features.csv	v1
item_features	item_id	item_interaction_count	int64	data\transformed\item_features.csv	v1
item_features	item_id	price	float64	data\transformed\item_features.csv	v1
item_features	item_id	price_norm	float64	data\transformed\item_features.csv	v1
item_features	item_id	title	object	data\transformed\item_features.csv	v1
item_features	item_id	unique_users	int64	data\transformed\item_features.csv	v1
user_features	user_id	activity_frequency	int64	data\transformed\user_features.csv	v1
user_features	user_id	avg_interaction_weight	float64	data\transformed\user_features.csv	v1
user_features	user_id	avg_rating_per_user	float64	data\transformed\user_features.csv	v1
user_features	user_id	unique_items_interacted	int64	data\transformed\user_features.csv	v1
user_item_features	user_id, item_id	addtocart	int64	data\transformed\user_item_features.csv	v1
user_item_features	user_id, item_id	interaction_count	int64	data\transformed\user_item_features.csv	v1
user_item_features	user_id, item_id	last_event_ts	float64	data\transformed\user_item_features.csv	v1
user_item_features	user_id, item_id	total_interaction_weight	float64	data\transformed\user_item_features.csv	v1
user_item_features	user_id, item_id	transaction	int64	data\transformed\user_item_features.csv	v1
user_item_features	user_id, item_id	view	int64	data\transformed\user_item_features.csv	v1

10. Sample Training Feature Retrieval

user_id	item_id	interaction_count	total_interaction_weight	last_event_ts	activity_frequency	unique_items_interacted	avg_rating_per_user	item_interaction_count	unique_users	avg_rating_per_item	category	price_norm
0	67045	1	1.0		3	3	1.0	173	170	1.046242774566474		
0	285930	1	1.0		3	3	1.0	74	70	1.108108108108108		
0	357564	1	1.0		3	3	1.0	248	245	1.0483870967741935		
1	72028	1	1.0		1	1	1.0	42	42	1.0		

user_id	item_id	interaction_count	total_interaction_weight	last_event_timestamps	activity_frequency	unique_items_interacted	avg_rating_per_user	item_interaction_count	unique_users	avg_rating_per_item	category	price_norm
2	216305	1	1.0		4	4	1.0	443	440	1.018058690744921		
2	259884	1	1.0		4	4	1.0	261	247	1.137931034482759		
2	325215	1	1.0		4	4	1.0	99	94	1.1414141414141414		
2	342816	1	1.0		4	4	1.0	87	77	1.3448275862068966		
3	385090	1	1.0		1	1	1.0	5	5	1.0		
4	177677	1	1.0		1	1	1.0	52	52	1.0		

11. Sample Inference Feature Retrieval

request_user_id	item_id	item_interaction_count	unique_users	avg_rating_per_item	category	price_norm
0	67045	173	170	1.046242774566474		
0	72028	42	42	1.0		
0	177677	52	52	1.0		
0	216305	443	440	1.018058690744921		
0	259884	261	247	1.137931034482759		
0	285930	74	70	1.108108108108108		
0	325215	99	94	1.1414141414141414		
0	342816	87	77	1.3448275862068966		
0	357564	248	245	1.0483870967741935		
0	385090	5	5	1.0		

12. Feature Store Folder Structure

feature_store/

```

feature_store/
├── docs/
│   ├── sample_inference_feature_retrieval.csv
│   └── sample_training_feature_retrieval.csv
├── offline_store/
│   └── feature_store.db
├── registry/
│   ├── feature_metadata.csv
│   ├── feature_metadata.json
│   ├── feature_registry.yaml
│   ├── feature_store_config.json
│   └── feature_store_schema.sql

```

data/transformed or data/features

```

transformed/
├── item_cooccurrence.csv
├── item_features.csv
├── user_features.csv
└── user_item_features.csv

```

13. Feature Store Configuration Preview

```
{
  "project": "RecoMart Feature Store",
  "store_type": "custom_metadata_registry",
  "version": "v1",
  "created_ts": "2026-04-29T16:46:23Z",
  "entities": [
    "user_id",
    "item_id"
  ],
  "offline_store": {
    "type": "sqlite",
    "path": "feature_store\\offline_store\\feature_store.db"
  },
  "registry": {
    "metadata_csv": "feature_store\\registry\\feature_metadata.csv",
    "metadata_json": "feature_store\\registry\\feature_metadata.json"
  },
  "feature_views": [
    {
      "name": "user_features",
      "version": "v1",
      "entities": [
        "user_id"
      ],
      "source_path": "data\\transformed\\user_features.csv",
      "feature_count": 4,
      "features": [
        "activity_frequency",
        "unique_items_interacted",
        "avg_interaction_weight",
        "avg_rating_per_user"
      ],
      "online_enabled": true,
      "offline_enabled": true
    },
    {
      "name": "item_features",
      "version": "v1",
      "entities": [
        "item_id"
      ],
      "source_path": "data\\transformed\\item_features.csv",
      "feature_count": 8,
      "features": [
        "item_interaction_count",
        "unique_users",
        "avg_interaction_weight",
        "avg_rating_per_item",
        "title",
        "category",
        "price",
        "price_norm"
      ],
      "online_enabled": true,
      "offline_enabled": true
    },
    {
      "name": "user_item_features",
      "version": "v1",
      "entities": [
        "user_id",
        "item_id"
      ],
      "source_path": "data\\transformed\\user_item_features.csv",
      "feature_count": 6,
      "features": [
        "interaction_count",
        "total_interaction_weight",
        "last_event_ts",
        "addtocart",
        "transaction",
        "view"
      ],
      "online_enabled": true,
      "offline_enabled": true
    },
    {
      "name": "item_cooccurrence",
      "version": "v1",
      "entities": [
        "item_id_a",
        "item_id_b"
      ],
      "source_path": "data\\transformed\\item_cooccurrence.csv",
      "feature_count": 1,
      "features": [
```

```

        "cooccurrence_count"
    ],
    "online_enabled": true,
    "offline_enabled": true
}
}
}

```

14. Feature Metadata JSON Preview

```

[
  {
    "feature_view": "item_cooccurrence",
    "entity_keys": "item_id_a, item_id_b",
    "feature_name": "cooccurrence_count",
    "dtype": "int64",
    "source_path": "data\\transformed\\item_cooccurrence.csv",
    "source_table": "item_cooccurrence",
    "transformation_logic": "Count of times two items appeared in the same user history.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "item_features",
    "entity_keys": "item_id",
    "feature_name": "avg_interaction_weight",
    "dtype": "float64",
    "source_path": "data\\transformed\\item_features.csv",
    "source_table": "item_features",
    "transformation_logic": "Mean interaction weight based on event type mapping.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "item_features",
    "entity_keys": "item_id",
    "feature_name": "avg_rating_per_item",
    "dtype": "float64",
    "source_path": "data\\transformed\\item_features.csv",
    "source_table": "item_features",
    "transformation_logic": "Average explicit rating or proxy weighted score per item.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "item_features",
    "entity_keys": "item_id",
    "feature_name": "category",
    "dtype": "object",
    "source_path": "data\\transformed\\item_features.csv",
    "source_table": "item_features",
    "transformation_logic": "Joined item category from product metadata.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "item_features",
    "entity_keys": "item_id",
    "feature_name": "item_interaction_count",
    "dtype": "int64",
    "source_path": "data\\transformed\\item_features.csv",
    "source_table": "item_features",
    "transformation_logic": "Count of interactions per item.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "item_features",
    "entity_keys": "item_id",
    "feature_name": "price",
    "dtype": "float64",
    "source_path": "data\\transformed\\item_features.csv",
    "source_table": "item_features",
    "transformation_logic": "Joined product price.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "item_features",
    "entity_keys": "item_id",

```



```

    "feature_name": "price_norm",
    "dtype": "float64",
    "source_path": "data\\transformed\\item_features.csv",
    "source_table": "item_features",
    "transformation_logic": "Normalized product price.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "item_features",
    "entity_keys": "item_id",
    "feature_name": "title",
    "dtype": "object",
    "source_path": "data\\transformed\\item_features.csv",
    "source_table": "item_features",
    "transformation_logic": "Joined item title from product metadata.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "item_features",
    "entity_keys": "item_id",
    "feature_name": "unique_users",
    "dtype": "int64",
    "source_path": "data\\transformed\\item_features.csv",
    "source_table": "item_features",
    "transformation_logic": "Distinct user count per item.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_features",
    "entity_keys": "user_id",
    "feature_name": "activity_frequency",
    "dtype": "int64",
    "source_path": "data\\transformed\\user_features.csv",
    "source_table": "user_features",
    "transformation_logic": "Count of interactions per user.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_features",
    "entity_keys": "user_id",
    "feature_name": "avg_interaction_weight",
    "dtype": "float64",
    "source_path": "data\\transformed\\user_features.csv",
    "source_table": "user_features",
    "transformation_logic": "Mean interaction weight based on event type mapping.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_features",
    "entity_keys": "user_id",
    "feature_name": "avg_rating_per_user",
    "dtype": "float64",
    "source_path": "data\\transformed\\user_features.csv",
    "source_table": "user_features",
    "transformation_logic": "Average explicit rating or proxy weighted score per user.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_features",
    "entity_keys": "user_id",
    "feature_name": "unique_items_interacted",
    "dtype": "int64",
    "source_path": "data\\transformed\\user_features.csv",
    "source_table": "user_features",
    "transformation_logic": "Distinct item count per user.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_item_features",
    "entity_keys": "user_id, item_id",
    "feature_name": "addtocart",
    "dtype": "int64",
    "source_path": "data\\transformed\\user_item_features.csv",
    "source_table": "user_item_features",
    "transformation_logic": "Count of add-to-cart events for a user-item pair.",

```

```

    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_item_features",
    "entity_keys": "user_id, item_id",
    "feature_name": "interaction_count",
    "dtype": "int64",
    "source_path": "data\\transformed\\user_item_features.csv",
    "source_table": "user_item_features",
    "transformation_logic": "Count of user-item interactions.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_item_features",
    "entity_keys": "user_id, item_id",
    "feature_name": "last_event_ts",
    "dtype": "float64",
    "source_path": "data\\transformed\\user_item_features.csv",
    "source_table": "user_item_features",
    "transformation_logic": "Most recent interaction timestamp for the user-item pair.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_item_features",
    "entity_keys": "user_id, item_id",
    "feature_name": "total_interaction_weight",
    "dtype": "float64",
    "source_path": "data\\transformed\\user_item_features.csv",
    "source_table": "user_item_features",
    "transformation_logic": "Sum of interaction weights for the user-item pair.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_item_features",
    "entity_keys": "user_id, item_id",
    "feature_name": "transaction",
    "dtype": "int64",
    "source_path": "data\\transformed\\user_item_features.csv",
    "source_table": "user_item_features",
    "transformation_logic": "Count of transaction events for a user-item pair.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  },
  {
    "feature_view": "user_item_features",
    "entity_keys": "user_id, item_id",
    "feature_name": "view",
    "dtype": "int64",
    "source_path": "data\\transformed\\user_item_features.csv",
    "source_table": "user_item_features",
    "transformation_logic": "Count of view events for a user-item pair.",
    "version": "v1",
    "retrieval_modes": "training,inference",
    "created_ts": "2026-04-29T16:46:23Z"
  }
]

```

15. Feature Store SQL Schema

```

CREATE TABLE "feature_metadata" (
  "feature_view" TEXT,
  "entity_keys" TEXT,
  "feature_name" TEXT,
  "dtype" TEXT,
  "source_path" TEXT,
  "source_table" TEXT,
  "transformation_logic" TEXT,
  "version" TEXT,
  "retrieval_modes" TEXT,
  "created_ts" TEXT
);

CREATE TABLE "feature_store_versions" (
  "version" TEXT,
  "created_ts" TEXT,
  "store_type" TEXT,
  "offline_store" TEXT,

```

```

    "config_path" TEXT,
    "metadata_path" TEXT
);

CREATE TABLE "item_cooccurrence_v1" (
    "item_id_a" INTEGER,
    "item_id_b" INTEGER,
    "cooccurrence_count" INTEGER
);

CREATE TABLE "item_features_v1" (
    "item_id" INTEGER,
    "item_interaction_count" INTEGER,
    "unique_users" INTEGER,
    "avg_interaction_weight" REAL,
    "avg_rating_per_item" REAL,
    "title" TEXT,
    "category" TEXT,
    "price" REAL,
    "price_norm" REAL
);

CREATE TABLE "user_features_v1" (
    "user_id" INTEGER,
    "activity_frequency" INTEGER,
    "unique_items_interacted" INTEGER,
    "avg_interaction_weight" REAL,
    "avg_rating_per_user" REAL
);

CREATE TABLE "user_item_features_v1" (
    "user_id" INTEGER,
    "item_id" INTEGER,
    "interaction_count" INTEGER,
    "total_interaction_weight" REAL,
    "last_event_ts" REAL,
    "addtocart" INTEGER,
    "transaction" INTEGER,
    "view" INTEGER
);

```

16. Feature Store Code / Asset Preview

Asset: *feature_store_config.json*

Path: feature_store\registry\feature_store_config.json

```
{
  "project": "RecoMart Feature Store",
  "store_type": "custom_metadata_registry",
  "version": "v1",
  "created_ts": "2026-04-29T16:46:23Z",
  "entities": [
    "user_id",
    "item_id"
  ],
  "offline_store": {
    "type": "sqlite",
    "path": "feature_store\\offline_store\\feature_store.db"
  },
  "registry": {
    "metadata_csv": "feature_store\\registry\\feature_metadata.csv",
    "metadata_json": "feature_store\\registry\\feature_metadata.json"
  },
  "feature_views": [
    {
      "name": "user_features",
      "version": "v1",
      "entities": [
        "user_id"
      ],
      "source_path": "data\\transformed\\user_features.csv",
      "feature_count": 4,
      "features": [
        "activity_frequency",
        "unique_items_interacted",
        "avg_interaction_weight",
        "avg_rating_per_user"
      ],
      "online_enabled": true,
      "offline_enabled": true
    },
    {
      "name": "item_features",
      "version": "v1",
      "entities": [
        "item_id"
      ],
      "source_path": "data\\transformed\\item_features.csv",
      "feature_count": 8,
      "features": [
        "item_interaction_count",
        "unique_users",
        "avg_interaction_weight",
        "avg_rating_per_item",
        "title",
        "category",
        "price",
        "price_norm"
      ],
      "online_enabled": true,
      "offline_enabled": true
    },
    {
      "name": "user_item_features",
      "version": "v1",
      "entities": [
        "user_id",
        "item_id"
      ],
      "source_path": "data\\transformed\\user_item_features.csv",
      "feature_count": 6,
      "features": [
        "interaction_count",
        "total_interaction_weight",
        "last_event_ts",
        "addtocart",
        "transaction",
        "view"
      ],
      "online_enabled": true,
      "offline_enabled": true
    },
    {
      "name": "item_cooccurrence",
      "version": "v1",
      "entities": [
        "item_id_a",

```

```

        "item_id_b"
    ],
    "source_path": "data\\transformed\\item_cooccurrence.csv",
    "feature_count": 1,
    "features": [
        "cooccurrence_count"
    ],
    "online_enabled": true,
    "offline_enabled": true
}
]
}

```

Asset: feature_store_schema.sql

Path: feature_store\registry\feature_store_schema.sql

```

CREATE TABLE "feature_metadata" (
    "feature_view" TEXT,
    "entity_keys" TEXT,
    "feature_name" TEXT,
    "dtype" TEXT,
    "source_path" TEXT,
    "source_table" TEXT,
    "transformation_logic" TEXT,
    "version" TEXT,
    "retrieval_modes" TEXT,
    "created_ts" TEXT
);

CREATE TABLE "feature_store_versions" (
    "version" TEXT,
    "created_ts" TEXT,
    "store_type" TEXT,
    "offline_store" TEXT,
    "config_path" TEXT,
    "metadata_path" TEXT
);

CREATE TABLE "item_cooccurrence_v1" (
    "item_id_a" INTEGER,
    "item_id_b" INTEGER,
    "cooccurrence_count" INTEGER
);

CREATE TABLE "item_features_v1" (
    "item_id" INTEGER,
    "item_interaction_count" INTEGER,
    "unique_users" INTEGER,
    "avg_interaction_weight" REAL,
    "avg_rating_per_item" REAL,
    "title" TEXT,
    "category" TEXT,
    "price" REAL,
    "price_norm" REAL
);

CREATE TABLE "user_features_v1" (
    "user_id" INTEGER,
    "activity_frequency" INTEGER,
    "unique_items_interacted" INTEGER,
    "avg_interaction_weight" REAL,
    "avg_rating_per_user" REAL
);

CREATE TABLE "user_item_features_v1" (
    "user_id" INTEGER,
    "item_id" INTEGER,
    "interaction_count" INTEGER,
    "total_interaction_weight" REAL,
    "last_event_ts" REAL,
    "addtocart" INTEGER,
    "transaction" INTEGER,
    "view" INTEGER
);

```

Asset: feature_store_20260429T072423Z.json

Path: logs\feature_store_20260429T072423Z.json

```

{
    "stage": "feature_store",
    "status": "success",
    "registry_path": "C:\\Users\\barath\\recomart-
pipeline\\data\\feature_registry\\feature_registry_20260429T072423Z.json"
}

```

```
}
```

Asset: registry_loader_failed_20260429_160933.json

Path: logs\monitoring\registry_loader_failed_20260429_160933.json

```
{
  "run_ts": "20260429_160933",
  "stage": "registry_loader",
  "status": "failed",
  "notebook_path": "C:\\Users\\barath\\recomart-
pipeline\\src\\04-features\\registry_loader.ipynb",
  "output_notebook": "C:\\Users\\barath\\recomart-pipeline\\runs\\prefect\\20260429_160933\\exe
cuted_notebooks\\registry_loader\\registry_loader_executed.ipynb",
  "error": "4.61s/cell\\nExecuting: 33%|\\u2588\\u2588\\u2588\\u258e      | 1/3 [00:05<00:10,
5.05s/cell]\\nTraceback (most recent call last):\\n File \\\"<frozen runpy>\\", line 198, in
_run_module_as_main\\n File \\\"<frozen runpy>\\", line 88, in _run_code\\n File
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\papermill\\__main__.py\\", line 4, in <module>\\n   papermill()\\n File
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\click\\core.py\\", line 1514, in __call__\\n   return self.main(*args, **kwargs)\\n
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\click\\core.py\\", line 1435, in main\\n   rv = self.invoke(ctx)\\n
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\click\\core.py\\", line 1298, in invoke\\n   return ctx.invoke(self.callback,
**ctx.params)\\n
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\click\\core.py\\", line 853, in invoke\\n   return callback(*args, **kwargs)\\n
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\click\\decorators.py\\", line 34, in new_func\\n   return f(get_current_context(),
*args, **kwargs)\\n
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\papermill\\cli.py\\", line 235, in papermill\\n   execute_notebook(\\n File
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\papermill\\execute.py\\", line 131, in execute_notebook\\n
raise_for_execution_errors(nb, output_path)\\n File
\\\"c:\\Users\\barath\\AppData\\Local\\anaconda3\\envs\\dm4ml\\Lib\\site-
packages\\papermill\\execute.py\\", line 251, in raise_for_execution_errors\\n   raise
error\\npapermill.exceptions.PapermillExecutionError:
\\n-----\\nException
encountered at \\\"In [1]\\":\\n-----\\n
-----\\nFileNotFoundError                                Traceback (most recent call last)\\nCell
In[1], line 367\\n   365 if __name__ == \"__main__\":\\n   366     print(Path.cwd())\\n--> 367
demo()\\n\\nCell In[1], line 305, in demo(registry_path, version)\\n   301 def demo(\\n
302     registry_path: Union[str, Path] = DEFAULT_REGISTRY_PATH,\\n   303     version:
Optional[str] = None\\n   304 ) -> None:\\n--> 305     registry =
load_feature_registry(registry_path)\\n   307     errors = validate_registry(registry)\\n   308
if errors:\\n\\nCell In[1], line 23, in load_feature_registry(registry_path)\\n   20
registry_path = Path(registry_path).resolve()\\n   22 if not registry_path.exists():\\n--> 23
raise FileNotFoundError(f\"Registry file not found: {registry_path}\\\")\\n   25 with
open(registry_path, \"r\", encoding=\"utf-8\") as f:\\n   26     registry =
yaml.safe_load(f)\\n\\nFileNotFoundError: Registry file not found:
C:\\Users\\barath\\feature_store\\registry\\feature_registry.yaml\\n\\n\",
  "event_ts": "2026-04-29T16:26:06.655592"
}
```

Asset: registry_loader_started_20260429_160933.json

Path: logs\monitoring\registry_loader_started_20260429_160933.json

```
{
  "run_ts": "20260429_160933",
  "stage": "registry_loader",
  "status": "started",
  "notebook_path": "C:\\Users\\barath\\recomart-
pipeline\\src\\04-features\\registry_loader.ipynb",
  "output_notebook": "C:\\Users\\barath\\recomart-pipeline\\runs\\prefect\\20260429_160933\\exe
cuted_notebooks\\registry_loader\\registry_loader_executed.ipynb",
  "error": null,
  "event_ts": "2026-04-29T16:26:00.194961"
}
```

17. Validation / Pipeline Log Preview

```
{
  "cells": [
    {
      "cell_type": "code",
      "execution_count": 2,
```

```

    "id": "fef32579-f890-4b53-8d24-2fcae64a4914",
    "metadata": {},
    "outputs": [
        {
            "name": "stdout",
            "output_type": "stream",
            "text": [
                "PROJECT_ROOT: C:\\Users\\barath\\recomart-pipeline\\n",
                "RAW_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\n",
                "BRONZE_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\bronze\\n",
                "\\n",
                "Discovered files:\\n",
                "- events_csv: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\events.csv\\n",
                "- category_tree_csv: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-26\\load_hour=19\\category_tree.csv\\n",
                "- item_properties_part1_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-28\\load_hour=14\\item_properties_part1.csv\\n",
                "- item_properties_part2_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-28\\load_hour=14\\item_properties_part2.csv\\n",
                "- products_raw_json: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\products_raw.json\\n",
                "- categories_raw_json: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\categories_raw.json\\n",
                "- products_parquet: C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\products.parquet\\n",
                "- categories_parquet: C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet\\n",
                "\\n",
                "Initial status: FAIL\\n",
                "Final status: PASS\\n",
                "Initial issues file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260428T144031Z.csv\\n",
                "Initial summary file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260428T144031Z.csv\\n",
                "Revalidation issues file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_revalidated_20260428T144031Z.csv\\n",
                "Revalidation summary file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_revalidated_20260428T144031Z.csv\\n",
                "Fix log file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\fix_log_20260428T144031Z.csv\\n",
                "JSON report file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.json\\n",
                "PDF report file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.pdf\\n"
            ]
        }
    ],
    "source": [
        "# DM4ML - Assignment - Validation + Auto-Fix + Revalidation + PDF Report\\n",
        "\\n",
        "import json\\n",
        "from pathlib import Path\\n",
        "from datetime import datetime, timezone\\n",
        "\\n",
        "import pandas as pd\\n",
        "\\n",
        "from reportlab.lib import colors\\n",
        "from reportlab.lib.pagesizes import A4\\n",
        "from reportlab.lib.styles import getSampleStyleSheet\\n",
        "from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle,
PageBreak\\n",
        "\\n",
        "# =====\\n",
        "# RecoMart Validation - robust file discovery + validation\\n",
        "# with remediation, revalidation, and PDF reporting\\n",
        "# =====\\n",
        "\\n",
        "PROJECT_ROOT = Path.cwd().resolve()\\n",
        "while not (PROJECT_ROOT / '\\src\\').exists() and PROJECT_ROOT != PROJECT_ROOT.parent:\\n",
        "    PROJECT_ROOT = PROJECT_ROOT.parent\\n",
        "\\n",
        "RAW_ROOT = PROJECT_ROOT / '\\data\\' / '\\raw\\'\\n",
        "BRONZE_ROOT = PROJECT_ROOT / '\\data\\' / '\\bronze\\'\\n",
        "\\n",
        "REMIEDIATED_RAW_ROOT = PROJECT_ROOT / '\\data\\' / '\\remediated\\' / '\\raw\\'\\n",
        "REMIEDIATED_BRONZE_ROOT = PROJECT_ROOT / '\\data\\' / '\\remediated\\' / '\\bronze\\'\\n",
        "\\n",
        "REPORT_ROOT = PROJECT_ROOT / '\\reports\\' / '\\validation\\'\\n",
        "LOG_ROOT = PROJECT_ROOT / '\\logs\\'\\n",
        "\\n",
        "for folder in [REPORT_ROOT, LOG_ROOT, REMEDIATED_RAW_ROOT, REMEDIATED_BRONZE_ROOT]:\\n",
        "    folder.mkdir(parents=True, exist_ok=True)\\n",
        "\\n",
        "UTC_NOW = datetime.now(timezone.utc)\\n",
        "RUN_ID = UTC_NOW.strftime('%Y%m%dT%H%M%S')\\n",
        "RUN_TS = UTC_NOW.isoformat()\\n",
        "\\n",
        "LOG_FILE = LOG_ROOT / f'validation_log_{RUN_ID}.json\\'\\n",

```

```
"INITIAL_ISSUES_FILE = REPORT_ROOT / f"validation_issues_initial_{RUN_ID}.csv"\n",
"INITIAL_SUMMARY_FILE = REPORT_ROOT / f"validation_summary_initial_{RUN_ID}.csv"\n",
```

18. Data Quality Report Preview

```
{
  "run_id": "20260429T111342Z",
  "run_ts": "2026-04-29T11:13:42.777701+00:00",
  "project_root": "C:\\Users\\barath\\recomart-pipeline",
  "initial_validation": {
    "run_id": "20260429T111342Z",
    "run_ts": "2026-04-29T11:13:42.777701+00:00",
    "phase": "initial",
    "project_root": "C:\\Users\\barath\\recomart-pipeline",
    "raw_root": "C:\\Users\\barath\\recomart-pipeline\\data\\raw",
    "bronze_root": "C:\\Users\\barath\\recomart-pipeline\\data\\bronze",
    "discovered_files": {
      "events_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\events.csv",
      "category_tree_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\category_tree.csv",
      "item_properties_part1_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part1.csv",
      "item_properties_part2_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part2.csv",
      "products_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\products_raw.json",
      "categories_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\categories_raw.json",
      "products_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\products.parquet",
      "categories_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet"
    },
    "overall_status": "FAIL",
    "issues_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260429T111342Z.csv",
    "summary_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260429T111342Z.csv",
    "datasets": [
      {
        "phase": "initial",
        "dataset": "retailrocket_events",
        "rows": 2756101,
        "columns": 5,
        "errors": 0,
        "warnings": 2734104,
        "status": "PASS"
      },
      {
        "phase": "initial",
        "dataset": "retailrocket_category_tree",
        "rows": 1669,
        "columns": 2,
        "errors": 0,
        "warnings": 25,
        "status": "PASS"
      },
      {
        "phase": "initial",
        "dataset": "retailrocket_item_properties",
        "rows": 20275902,
        "columns": 4,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
      },
      {
        "phase": "initial",
        "dataset": "dummyjson_products_raw",
        "rows": 194,
        "columns": 27,
        "errors": 0,
        "warnings": 92,
        "status": "PASS"
      },
      {
        "phase": "initial",
        "dataset": "dummyjson_categories_raw",
        "rows": 24,
        "columns": 3,
        "errors": 1,
        "warnings": 0,
        "status": "FAIL"
      },
      {
        "phase": "initial",

```



```

        "dataset": "products_parquet",
        "rows": 194,
        "columns": 32,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
    },
    {
        "phase": "initial",
        "dataset": "categories_parquet",
        "rows": 24,
        "columns": 8,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
    }
]
},
"revalidation": {
    "run_id": "20260429T111342Z",
    "run_ts": "2026-04-29T11:13:42.777701+00:00",
    "phase": "revalidated",
    "remediated_raw_root": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw",
    "remediated_bronze_root": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\bronze",
    "discovered_files": {
        "events_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw\\events.csv",
        "category_tree_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\category_tree.csv",
        "item_properties_part1_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\item_properties_part1.csv",
        "item_properties_part2_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\item_properties_part2.csv",
        "products_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\products_raw.json",
        "categories_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\categories_raw.json",
        "products_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\bronze\\products.parquet",
        "categories_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\bronze\\categories.parquet"
    },
    "overall_status": "PASS",
    "issues_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_revalidated_20260429T111342Z.csv",
    "summary_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_revalidated_20260429T111342Z.csv",
    "fix_log_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\fix_log_20260429T111342Z.csv",
    "datasets": [
        {
            "phase": "revalidated",
            "dataset": "retailrocket_events",
            "rows": 2755641,
            "columns": 5,
            "errors": 0,
            "warnings": 2733184,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "retailrocket_category_tree",
            "rows": 1669,
            "columns": 2,
            "errors": 0,
            "warnings": 25,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "retailrocket_item_properties",
            "rows": 20275902,
            "columns": 4,
            "errors": 0,
            "warnings": 0,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "dummyjson_products_raw",
            "rows": 194,
            "columns": 27,
            "errors": 0,
            "warnings": 92,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "dummyjson_categories_raw",
            "rows": 0,

```

```

        "columns": 1,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
    },
    {
        "phase": "revalidated",
        "dataset": "products_parquet",
        "rows": 194,
        "columns": 32,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
    },
    {
        "phase": "revalidated",
        "dataset": "categories_parquet",
        "rows": 24,
        "columns": 8,
        "errors": 0,
        "warnings": 0,
        "status": "PASS"
    }
]
},
{
    "pdf_report_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260429T111342Z.pdf"
}

```

19. Validation Event Log Preview

```

{"event_ts": "2026-04-29T11:13:42.785703+00:00", "stage": "validation", "status": "started",
"message": "Validation started", "run_id": "20260429T111342Z"}
{"event_ts": "2026-04-29T11:14:14.271184+00:00", "stage": "remediation", "status": "started",
"message": "Validation failed; starting remediation", "run_id": "20260429T111342Z"}
{"event_ts": "2026-04-29T11:17:44.036779+00:00", "stage": "validation", "status": "completed",
"message": "Validation workflow completed", "run_id": "20260429T111342Z", "initial_status":
"FAIL", "final_status": "PASS", "pdf_report_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260429T111342Z.pdf"}

```

20. Feature Store Event Log Preview

```

{"event_ts": "2026-04-29T16:46:23.563551+00:00", "stage": "pipeline", "status": "started",
"message": "Feature store notebook execution started", "extra": {}}
{"event_ts": "2026-04-29T16:46:23.579594+00:00", "stage": "feature_source", "status":
"started", "message": "Discovering feature source tables", "extra": {}}
{"event_ts": "2026-04-29T16:46:31.944407+00:00", "stage": "feature_source", "status":
"completed", "message": "Loaded existing transformed feature tables", "extra": {"db_candidate":
"warehouse\\recommendation_features.db", "paths": {"user_features":
"data\\transformed\\user_features.csv", "item_features":
"data\\transformed\\item_features.csv", "user_item_features":
"data\\transformed\\user_item_features.csv", "item_cooccurrence":
"data\\transformed\\item_cooccurrence.csv"}}}
{"event_ts": "2026-04-29T16:46:31.961448+00:00", "stage": "feature_store", "status": "started",
"message": "Writing feature store artifacts", "extra": {}}
{"event_ts": "2026-04-29T16:47:42.369722+00:00", "stage": "feature_store", "status":
"completed", "message": "Feature store artifacts written", "extra": {"db":
"feature_store\\offline_store\\feature_store.db", "config_json":
"feature_store\\registry\\feature_store_config.json", "metadata_csv":
"feature_store\\registry\\feature_metadata.csv", "metadata_json":
"feature_store\\registry\\feature_metadata.json"}}
{"event_ts": "2026-04-29T16:47:43.619873+00:00", "stage": "pipeline", "status": "completed",
"message": "Feature store notebook execution completed", "extra": {"output_pdf": "07 Feature
Store- DM4ML-Group51.pdf", "feature_store_db":
"feature_store\\offline_store\\feature_store.db", "feature_metadata_csv":
"feature_store\\registry\\feature_metadata.csv"}}

```

21. Conclusion

This PDF consolidates the feature store deliverables for the assignment, including a simple versioned feature store implementation, feature metadata documentation, and sample training and inference retrieval demonstrations.